30-Minute Student Task

Build Your Own Rule-Based Python Chatbot

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Programming Language	Python

Task:

Build a simple rule-based conversational chatbot in Python. Your chatbot must interact continuously with the user, recognize different messages, provide appropriate responses, and allow the user to exit

Demonstration Evidence:

The following pages contain screenshots of the Python implementation and the successful terminal execution..

Python Chatbot Implementation

Screenshot of the completed Python program demonstrating the required rule-based conversation logic

```python
# Simple Python Chatbot

name = input("Hi! What is your name? ")

print("Hello", name + "! Welcome to my chatbot.")
print("You can ask me about the weather, Python, or my purpose.")
print("Type 'bye' or 'exit' to end the chat.")

while True:
    user_input = input("\nYou: ").lower().strip()

    # Greeting
    if "hello" in user_input or "hi" in user_input:
        print("Bot: Hello", name + "! How can I help you?")

    # Question 1
    elif "weather" in user_input:
        print("Bot: I cannot check live weather, but you can use a weather app.")

    # Question 2
    elif "python" in user_input:
        print("Bot: Python is a simple and popular programming language.")

    # Question 3
    elif "your name" in user_input or "who are you" in user_input:
        print("Bot: I am a simple Python chatbot created to answer your questions.")

    # Exit
    elif "bye" in user_input or "exit" in user_input:
        print("Bot: Goodbye", name + "! Have a nice day!")
        break

    # Unknown message
    else:
        print("Bot: Sorry, I don't understand that. Try asking about Python or weather.")
```

Chatbot Execution and Test Output

The output showing the chatbot responding to greetings, questions an unknown message, and the exit command.

```
Hi! What is your name? Priya
Hello Priya! Welcome to my chatbot.
You can ask me about the weather, Python, or my purpose.
Type 'bye' or 'exit' to end the chat.

You: Hello
Bot: Hello Priya! How can I help you?

You: What is python
Bot: Python is a simple and popular programming language.

You: What is the weather
Bot: I cannot check live weather, but you can use a weather app.

You: good morning
Bot: Sorry, I don't understand that. Try asking about Python or weather.

You: bye
Bot: Goodbye Priya! Have a nice day!
```

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